HW # 12

10.1

P 1 (a) Free energy of Van der Waal Gas:

$$F = -N\tau \left\{ \log\left[n_Q (V - Nb)/N \right] + 1 \right\} - N^2 a/V \longrightarrow \text{Eqn } 10.38$$

where $n_Q = \left(M\tau / 2\pi\hbar^2 \right)^{3/2} \longrightarrow$ Page 162.

Entropy $\rightarrow$ $\sigma = -\left.\frac{\partial F}{\partial \tau}\right)_V \longrightarrow$ Eqn. 5.63

$$F = -N\tau \left\{ \log\left[\left(\frac{M}{2\pi\hbar^2}\right)^{3/2} \frac{(V-Nb)}{N} \right] + \log \tau^{3/2} + 1 \right\} - N^2 a/V$$

Note I have used $\log(ab) = \log(a) + \log(b)$

$$\sigma = -\left.\frac{\partial F}{\partial \tau}\right)_V = -\left[-N\left\{ \log\left[\left(\frac{M}{2\pi\hbar^2}\right)^{3/2} \left(\frac{V-Nb}{N}\right) \right] + \log \tau^{3/2} + 1 \right\} - N\tau \left\{ \frac{3}{2}\frac{1}{\tau} \right\} \right]$$

$$= N \left\{ \log\left[n_Q \frac{(V-Nb)}{N} \right] + 5/2 \right\}$$

(b) use $U = F + \tau\sigma$

$$= -N\tau \left\{ \log\left[n_Q \frac{(V-Nb)}{N} \right] + 1 \right\} - \frac{N^2 a}{V} + N\tau \left\{ \log\left[\frac{n_Q (V-Nb)}{N} \right] + \frac{5}{2} \right\}$$

$$= \frac{3}{2} N\tau - \frac{N^2 a}{V}$$

(c) $\left(p + N^2 a/V^2 \right)(V - Nb) = N\tau$

$$\Rightarrow p(V - Nb) = N\tau - \frac{N^2 a}{V^2}(V - Nb)$$

$$\Rightarrow pV = \frac{\left[N\tau - \frac{N^2 a}{V^2}(V - Nb) \right] V}{V - Nb}$$

$$\Rightarrow \quad pV = \frac{\left[N\tau - \frac{N^2 a}{V^2}(V - Nb)\right]\cancel{V}}{\cancel{V}\left(1 - \frac{Nb}{V}\right)}$$

$$\frac{1}{1 - \frac{Nb}{V}} \approx 1 + \frac{Nb}{V} \quad ; \text{ since } \frac{Nb}{V} << 1$$

$$pV \approx \left[N\tau - \frac{N^2 a}{V^2}(V - Nb)\right]\left[1 + \frac{Nb}{V}\right]$$

all terms of the kind → a^2, ab, b^2 can be neglected

$$\Rightarrow pV \approx N\tau\left(1 + \frac{Nb}{V}\right) - \frac{N^2 a}{V}$$

Plugging the value of U & pV in
$H = U + pV$

we obtain

$$H = \frac{5}{2}N\tau + N^2 b\tau/V - 2N^2 a/V$$ → upto first order in a & b

let us express V in terms of p.

Consider → $pV \approx N\tau\left(1 + \frac{Nb}{V}\right) - \frac{N^2 a}{V}$

since a & b are small & one can approximate V as →

$$pV \approx N\tau \Rightarrow V \approx \frac{N\tau}{p} \Rightarrow \frac{1}{V} \approx \frac{p}{N\tau}$$

Note: further accuracy of V is obtained in powers of a & b as follows →

$$\frac{1}{V} \approx \frac{p}{N\tau} + a\, f(p, N, \tau) + b\, g(p, N, \tau) + \cdots$$

where f & g are some

functions of p, N and τ →

let us plug $\frac{1}{V} \approx \frac{p}{N\tau} + a\, f(p,N,\tau) + b\, g(p,N,\tau) + \cdots$

into → $H = \frac{5}{2} N\tau + \frac{N^2 b\tau}{V} - \frac{2N^2 a}{V}$

$$= \frac{5}{2}N\tau + N^2 b\tau\left[\frac{p}{N\tau} + a f(p,N,\tau) + b g(p,N,\tau) + \cdots\right] - 2N^2 a\left[\frac{p}{N\tau} + a f(p,N,\tau) + b g(p,N,\tau) + \cdots\right]$$

$$= \frac{5}{2}N\tau + Nbp - \frac{2Nap}{\tau} + \underbrace{\left(N^2 b\tau - 2N^2 a\right)\left(a f(p,N,\tau) + b g(p,N,\tau) + \cdots\right)}$$

Higher orders in a & b of the kind ab, a^2, b^2 which can be neglected!

Thus to linear order in a & b

$$H(\tau, p) = \frac{5}{2}N\tau + Nbp - 2Nap/\tau$$

10.4

P2

In example (26) we had →

$$Z_s = \sum_n \exp[-(n\hbar\omega - \varepsilon_0)/\tau] = \exp(+\varepsilon_0/\tau) \overbrace{\sum_n \exp\left(-n\frac{\hbar\omega}{\tau}\right)}^{\text{Geometric series.}}$$

For 3D

$$Z_s = \sum_{n_x, n_y, n_z} \exp\left\{-\left[(n_x+n_y+n_z)\hbar\omega - \varepsilon_0\right]/\tau\right\}$$

$$= \exp[\varepsilon_0/\tau] \sum_{n_x, n_y, n_z} \exp\left[-n_x\frac{\hbar\omega}{\tau}\right] \exp\left[-n_y\frac{\hbar\omega}{\tau}\right] \exp\left[-n_z\frac{\hbar\omega}{\tau}\right]$$

$$= \exp[\varepsilon_0/\tau] \left(\sum_{n_x} \exp\left[-n_x\frac{\hbar\omega}{\tau}\right]\right)\left(\sum_{n_y} \exp\left[-n_y\frac{\hbar\omega}{\tau}\right]\right)\left(\sum_{n_z} \exp\left(-n_z\frac{\hbar\omega}{\tau}\right)\right)$$

Geometric Series!

$$= \exp(\varepsilon_0/\tau) \left[\frac{1}{1-\exp(-\hbar\omega/\tau)}\right]^3$$

Vapor pressure is obtained from the condition:

$$\lambda_g = \lambda_s$$

where $\lambda_s \simeq \frac{1}{Z_s}$ → Eqn (29)

$$= \exp[-\varepsilon_0/\tau]\left[1-\exp(-\hbar\omega/\tau)\right]^3 \approx \exp(-\varepsilon_0/\tau)\left(\frac{\hbar\omega}{\tau}\right)^3$$

where since $\hbar\omega/\tau \ll 1 \rightarrow 1-\exp(-\hbar\omega/\tau) \approx \frac{\hbar\omega}{\tau}$.

$$\lambda_g = \frac{p}{\tau}\left(\frac{2m\hbar^2}{M\tau}\right)^{3/2} \longrightarrow \text{Eqn } 30$$

thus $$p \simeq \left(\frac{M}{2\pi}\right)^{3/2} \frac{\omega^3}{\tau^{1/2}} \exp(-\varepsilon_0/\tau)$$

b

$$L = H_g - H_s$$

for gas $H_g = U + pV$

at high temperatures $\rightarrow U \approx \frac{3}{2}\tau$

$pV = \tau$

$$\Rightarrow H_g = \frac{5}{2}\tau$$

for solid $\rightarrow$ $H_s \approx -\varepsilon_0 + 3\tau \leftarrow$ for a three-dimensional oscillator.

Groundstate energy

$$\Rightarrow L = \varepsilon_0 - \tau/2.$$

(a) Magnons are bosons, their number is not fixed (thus $\mu=0$)

$$\varepsilon_p = \frac{p^2}{2m}$$

$\Rightarrow$ number of magnons at temperature τ

$$N_m = \sum_p \frac{1}{e^{\frac{\varepsilon_p}{\tau}}-1} = \frac{V}{(2\pi\hbar)^3}\int d^3\vec{p}\,\frac{1}{e^{\varepsilon_p/\tau}-1}$$

Via density of states: $d^3\vec{p} = 4\pi\, p^2 dp = 4\pi\,(2m\varepsilon)\, d\sqrt{2m\varepsilon}$
$= 4\pi\,\frac{1}{2}(2m)^{3/2}\sqrt{\varepsilon}\, d\varepsilon$

$$\Rightarrow N_m = \frac{V}{(2\pi\hbar)^3}\, 2\pi\,(2m)^{3/2}\int_0^\infty \frac{d\varepsilon\,\sqrt{\varepsilon}}{e^{\varepsilon/\tau}-1} = (\varepsilon = x\tau) =$$

$$= \frac{V}{(2\pi\hbar)^3}\, 2\pi\,(2m\tau)^{3/2}\int_0^\infty \frac{dx\sqrt{x}}{e^x-1} = 2\pi V\left(\frac{2m\tau}{4\pi^2\hbar^2}\right)^{3/2} I_1$$

$$I_1 = \int_0^\infty dx\,\frac{\sqrt{x}}{e^x-1} = 2.315$$

(b) Total magnetization at $\tau=0$ is $M(0) = 2\mu_B N$, where N = number of spins.

Each magnon reduces magnetization by $2\mu_B$, hence total reduction of magnetization is $2\mu_B N_m$

$\Rightarrow$ we can write that fractional reduction is

$$\frac{M(0)-M(\tau)}{M(0)} = \frac{2\mu_B N_m}{2\mu_B N} = 2\pi\,\frac{V}{N}\left(\frac{2m\tau}{4\pi^2\hbar^2}\right)^{3/2} I_1$$

Note: if maximum magnetization is $M(0)$, and magnetization at τ is $M(\tau)$, then <u>reduction</u> of magnetization is $M(0)-M(\tau)$.

$$\frac{M(0)-M(\tau)}{M(0)} = 2\pi I_1 \frac{V}{N}\left(\frac{2m\tau}{h^2}\right)^{3/2} = \left(\frac{\tau}{\tau_0}\right)^{3/2}$$

$$\Rightarrow \tau_0 = \frac{h^2}{2m}\left(\frac{N}{V}\right)^{2/3}\frac{1}{(2\pi I_1)^{2/3}} = 4150\,K$$

Note that this can be re-written as

$$1-\frac{M(\tau)}{M(0)} = \left(\frac{\tau}{\tau_0}\right)^{3/2} \Rightarrow M(\tau) = M(0)\left[1-\left(\frac{\tau}{\tau_0}\right)^{3/2}\right]$$

which is known as Bloch's law

(c) $C_V = \left(\frac{\partial U}{\partial \tau}\right)_V \Rightarrow$ calculate $U(\tau)$ first.

$$U(\tau) = \sum_p \frac{\varepsilon_p}{e^{\frac{\varepsilon_p}{\tau}}-1} = \frac{V}{(2\pi\hbar)^3}\, 2\pi\,(2m)^{3/2}\int_0^\infty d\varepsilon\, \frac{\sqrt{\varepsilon}\cdot\varepsilon}{e^{\varepsilon/\tau}-1} =$$

$$= 2\pi\, V\left(\frac{2m}{h^2}\right)^{3/2}\tau^{5/2}\underbrace{\int_0^\infty dx\,\frac{x^{3/2}}{e^x-1}}_{I_2 = 1.783}$$

$$C_V = \frac{\partial U}{\partial \tau} = 5\pi V\left(\frac{2m\tau}{h^2}\right)^{3/2} I_2 = N\left(\frac{\tau}{\tau_1}\right)^{3/2}$$

where $\tau_1 = \frac{h^2}{2m}\left(\frac{N}{V}\right)^{2/3}\frac{1}{(5\pi I_2)^{2/3}} = 0.646\,\tau_0 \approx 2700\,K.$

Phonon contribution scales as $\left(\frac{\tau}{\theta_D}\right)^3$, θ_D = Debye temperature $= 470\,K$.

$\Rightarrow$ at $\tau > \theta_D$ magnon contribution to C_V is negligible, but at $\tau \ll \theta_D$ becomes important (in fact, below 2 K).

(d) For a two-dimensional array

$$N_m = \frac{A}{(2\pi\hbar)^2} \int d^2\vec{p}\, \frac{1}{e^{\varepsilon_p/\tau} - 1} \quad , \quad A = \text{area}$$

$$= \left(\frac{A}{2\pi\hbar}\right)^2 (2\pi) \int_0^\infty dp\, p\, \frac{1}{e^{\frac{p^2}{2m\tau}} - 1} =$$

$$= \frac{A}{(2\pi\hbar)^2} (2\pi m\tau) \int_0^\infty d\left(\frac{p^2}{2m\tau}\right) \frac{1}{e^{\frac{p^2}{2m\tau}} - 1} =$$

$$= \frac{A}{(2\pi\hbar)^2} (2\pi m\tau) \underbrace{\int_0^\infty \frac{dx}{e^x - 1}}_{I_3}$$

But the Integral I_3 diverges at $x \to 0$:

for small $x < x_0$

$$\int_0^{x_0} \frac{dx}{(1 + x + \frac{1}{2}x^2 + \dots) - 1} + \int_{x_0}^\infty \frac{dx}{e^x - 1} =$$

$$= \underbrace{\int_0^{x_0} \frac{dx}{x(1 + \frac{1}{2}x + \dots)}}_{\downarrow} + \int_{x_0}^\infty \frac{dx}{e^x - 1}$$

$$\int_0^{x_0} \frac{dx}{x} = \ln x \Big|_0^{x_0} = \ln\left(\frac{x_0}{0}\right) = +\infty.$$

$\Rightarrow$ the number of magnons diverges $\Rightarrow$ at finite τ, $M(\tau) = 0$!